BSc 2024/2025

For internal students of Imperial College.

Mathematics Foundations (BUSI40001)

3 hours

XXXday ABth January 2024, 09:00 – 12:00

Instructions for the exam

- For Section A, internet access with ChatGPT is allowed and a computer with R or Python is needed.
- Section B is closed-book and paper-based. Section B is to made accessible to the students after completing Section A.
- Answer ALL questions (The mark breakdown is indicated throughout the exam).
- It is important to write neatly and clearly.
- If a question or part of a question is unclear to you, state any assumptions you need in order to complete the question.

©Imperial College London 2024/2024

SECTION A (35 marks)

You have 45 minutes for Section A. Answer all **TWO** questions. Please submit your code. You must close your laptop after 45 minutes.

1. (20 marks) The ages at death of male parents and their male children are shown in the table below.

Person	A	B	C	D	E	F	G
adult	75	74	85	40	69	83	65
child	84	82	93	75	77	90	73

This question is about a male's lifespan given his father's lifespan.

(a) Write out the linear regression in the matrix form

$$\underline{y} = \mathbf{X}\underline{\beta} + \underline{\epsilon},$$

where $\underline{y}$, $\mathbf{X}$, $\underline{\beta}$, and $\underline{\epsilon}$ are to be defined. [2 marks]

[1/2 a mark for each vector]

$$\underline{x} = (75, 74, 85, 40, 69, 83, 65)^\top$$

is the 7×1 vector of male parent lifespans,

$$\underline{y} = (84, 82, 93, 75, 77, 90, 73)^\top$$

is the 7×1 vector of male child lifespans, $\underline{e} = (1, \ldots, 1)^\top$ is 7×1 vector of ones.

$$\mathbf{X} = [\underline{e}, \underline{x}]$$

is the 7 by 2 design matrix,

$$\underline{\beta} = [\beta_1, \beta_2]^\top$$

is the 1 by 2 parameter vector, and $\underline{\epsilon}$ is 7×1 error vector.

(b) Show that the expression for $\underline{\beta}$ which minimizes $\underline{\epsilon}^\top \underline{\epsilon}$ is given by

$$\widehat{\underline{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\underline{y}.$$

[5 marks]

Observe that $\underline{\epsilon}^\top\underline{\epsilon}$, where $\underline{\epsilon} = \underline{y} - \mathbf{X}\underline{\beta}$, is minimized by projecting $\underline{y}$ orthogonally into the column space of $\mathbf{X}$. Therefore, we want to choose $\underline{\beta}$ such that $\mathbf{X}^\top(\underline{y} - \mathbf{X}\underline{\beta}) = \underline{0}$. We shall denote the parameter vector which satisfies the latter equation via $\widehat{\underline{\beta}}$ and so

$$\mathbf{X}^\top(\underline{y} - \mathbf{X}\widehat{\underline{\beta}}) = \underline{0}$$

Therefore,

$$\mathbf{X}^\top\underline{y} = \mathbf{X}^\top\mathbf{X}\widehat{\underline{\beta}}.$$

Now $\mathbf{X}^\top\mathbf{X}$ is a 2 by 2 square matrix. Now

$$\widehat{\underline{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underline{y}.$$

(c) Show that the fitted values are given by

$$\underline{\hat{y}} = \mathbf{H}\underline{y}$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T$$

[3 marks]

The fitted values of the dependent variable are given by $\underline{\hat{y}} = \underline{\widehat{\beta}}\mathbf{X}$. Therefore,

$$\underline{\hat{y}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underline{y},$$

and so

$$\underline{\hat{y}} = \mathbf{H}\underline{y},$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T.$$

(d) Show that $\mathbf{H}$ is idempotent and symmetric. [5 marks]

$$\begin{aligned}
\mathbf{H}^2 &= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is idempotent. Now,

$$\begin{aligned}
\mathbf{H}^\top &= [\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T]^\top \\
&= [\mathbf{X}^\top]^\top[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top\mathbf{X}^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^\top]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top(\mathbf{X}^T)^\top\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is symmetric.

(e) Estimate the lifetime of a male whose male parent dies at age 89. You should do this using either R or Python [5 marks]

You need a calculator or a computer!

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 7 & 491 \\ 491 & 35801 \end{pmatrix}$$

$$\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1} = \begin{pmatrix} \frac{35801}{9526} & -\frac{491}{9526} \\ -\frac{491}{9526} & \frac{7}{9526} \end{pmatrix}$$

$$\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top = \begin{pmatrix} -\frac{512}{4763} & -\frac{533}{9526} & -\frac{2967}{4763} & \frac{16161}{9526} & \frac{961}{4763} & -\frac{2476}{4763} & \frac{1943}{4763} \\ \frac{17}{4763} & \frac{27}{9526} & \frac{52}{4763} & -\frac{211}{9526} & -\frac{4}{4763} & \frac{45}{4763} & -\frac{18}{4763} \end{pmatrix}$$

$$\begin{aligned} \underline{\widehat{\beta}} &= \left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top\underline{y} \\ &= \begin{pmatrix} \frac{46953}{866} \\ \frac{343}{866} \end{pmatrix} \\ &= \begin{pmatrix} 54.2182 \\ 0.396074 \end{pmatrix} \end{aligned}$$

$$54.2182 + 0.396074 \times 89 = 89.4688$$

Estimate of age at death for male child is 89.4688.

2. (15 marks: 5 marks for explanation, 10 marks for code)

Find and classify the critical points of the function $f(x, y) = x^2 + y^3 + xy$. Use code written in R or Python to carry out the necessary calculations, but explain the steps needed without showing algebraic details.

To find the critical points, we first compute the first-order partial derivatives of f with respect to x and y, set them equal to zero, and solve for x and y.

$$\frac{\partial f}{\partial x} = 0$$
$$\frac{\partial f}{\partial y} = 0$$

[1 mark]

So one critical point is at $(0, 0)$.

[1 mark]

The other critical point is at $\left(-\frac{1}{12}, \frac{1}{6}\right)$.

[1 mark]

We compute the second-order partial derivatives and use them to find the determinant of the Hessian.

The determinant D of the Hessian matrix is given by:

$$D = f_{xx}f_{yy} - (f_{xy})^2$$

- At $(0, 0)$: $D = -1$. Since $D < 0$, the critical point at $(0, 0)$ is a saddle point.

[1 mark]

At $\left(-\frac{1}{12}, \frac{1}{6}\right)$, $D > 0$ and $f_{xx} > 0$, the critical point is a local minimum.

[2 marks]

```
import sympy as sp

# Define symbols
x, y = sp.symbols('x y')

# Define the function
f = x**2 + y**3 + x*y

# Compute first-order partial derivatives
df_dx = sp.diff(f, x)
df_dy = sp.diff(f, y)

# Solve the system of equations df/dx = 0 and df/dy = 0
critical_points = sp.solve([df_dx, df_dy], (x, y))

# Compute second-order partial derivatives
f_xx = sp.diff(df_dx, x)
f_yy = sp.diff(df_dy, y)
f_xy = sp.diff(df_dx, y)

# Function to classify critical points
def classify_critical_point(point):
    x0, y0 = point
    D = f_xx.subs({x: x0, y: y0}) * f_yy.subs({x: x0, y: y0}) -
    f_xy.subs({x: x0, y: y0})**2
    f_xx_val = f_xx.subs({x: x0, y: y0})
    if D > 0:
        if f_xx_val > 0:
            classification = 'Local minimum'
        else:
            classification = 'Local maximum'
    elif D < 0:
        classification = 'Saddle point'
    else:
        classification = 'Indeterminate'
    return classification

# Output the results
for point in critical_points:
    classification = classify_critical_point(point)
    print(f"Critical point at {point}: {classification}")


Critical point at (0, 0): Saddle point
Critical point at (-1/12, 1/6): Local minimum
```

SECTION B (80 marks)

You have 2 hours 15 minutes for Section B. Answer all **FOUR** questions. Each question is worth 20 marks.

1. (20 marks) Consider a real sequence $(a_n)_n$

(a) Define what is meant by a subsequence of $(a_n)_n$. [3 marks]

Let (a_n) be a sequence of real numbers, and let $n_1 < n_2 < n_3 < n_4 < n_5 < \ldots$ be a strictly increasing sequence of natural numbers. Then the sequence

$$(a_{n_1}, a_{n_2}, a_{n_3}, a_{n_4}, a_{n_5}, \ldots)$$

is called a subsequence of (a_n) and is denoted by (a_{n_k}), where $k \in \mathbb{N}$ indexes the subsequence.

(b) Prove that a sequence converges to a limit if and only if every subsequence converges to that same limit, i.e.

$$a_n \to a \iff (\text{for every subsequence } a_{n_k}) a_{n_k} \to a$$

[4 marks]

We start by assuming $\lim a_n = a$, i.e.

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|a_n - a| < \epsilon.$$

Now, for every $m : \mathbb{N} \to \mathbb{N}$ such that $m(n+1) \geq m(n)$, we can find an M such that $m(M) > N$. Therefore,

$$(\forall \epsilon > 0)(\exists M \in \mathbb{N})(\forall n \geq M)|a(m(n)) - a| < \epsilon.$$

Hence, $a_{m_n} \to a$.
Now suppose that for every $m : \mathbb{N} \to \mathbb{N}$ such that $m(n+1) \geq m(n)$,

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|a(m(n)) - a| < \epsilon.$$

If this is true for every $m : \mathbb{N} \to \mathbb{N}$ such that $m(n+1) \geq m(n)$, we can just choose $m(n) = n$, so that

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|a_n - a| < \epsilon.$$

(c) What is meant by the peak of a sequence $(a_n)_n$? [1 mark]

a_m is a peak of the sequence (a_n) if $(\forall n > m) a_m \geq a_n$.

(d) Prove that every sequence contains a monotone subsequence. [8 marks]

A sequence will either have a fixed number of peaks, say $N \in \mathbb{N}$ or the number of peaks, like the natural numbers, will never end.
If the number of peaks never ends, let a_{n_k} be the k'th peak. Then for every $k \in \mathbb{N}$, we have $a_{n_k} \geq a_{n_{k+1}}$. Hence, (a_{n_k}) is a monotone decreasing subsequence.
Now, we consider the case where there are fixed numbers of peaks. Without loss of generality, ket a_K be the final peak. Now, $(\forall n \geq K) a_n \leq a_K$. Define $a_{n_1} = a_{K+1}$ and let a_{n_2} the the next term in the sequence which is greater than or equal to a_{n_1}. Such a term must exist, because a_{n_1} is not a peak. Similarly, let a_{n_3} the the next term in

the sequence which is greater than or equal to a_{n_2}, which must exist, because a_{n_2} is not a peak. Continuing in this way, we construct a monotone increasing subsequence.

(e) State the Monotone Convergence Theorem without proof. [1 mark]

If a sequence is monotone and bounded, then it converges.

(f) State and prove the Bolzano-Weierstrass Theorem. You may use results you have proved or stated earlier in this question. [3 marks]

Bolzano-Weierstrass Theorem

Every bounded sequence contains a convergent subsequence.

We have show that every sequence contains a monotone subsequence. Therefore, every bounded sequence contains a bounded and monotone subsequence, which by the Monotone Convergence Theorem, must converge.

2. (20 marks)

(a) Is the set of integers under the usual operations a field? Explain your answer. [2 marks]

No. Integers (other than 1 and -1) do not have multiplicative inverses within the integers, so they do not form a field.

(b) Prove that in a field $\mathbb{F}$, the additive identity satisfies the following property

$$(\forall x \in \mathbb{F}) x \times 0 = 0.$$

[8 marks]

In a field $\mathbb{F}$, the additive identity satisfies the following property

$$(\forall x \in \mathbb{F}) x \times 0 = 0.$$

- Start with the Distributive Law

$$(\forall a, b, c \in \mathbb{F}) a \times (b + c) = (a \times b) + (a \times c).$$

with $a = x$, $b = 0$ and $c = 0$, and so

$$x \times (0 + 0) = (x \times 0) + (x \times 0)$$

- We now use the Existence of Additive Identity

$$x + 0 = x$$

to obtain $0 + 0 = 0$, and so

$$x \times 0 = (x \times 0) + (x \times 0)$$

- We now use Commutativity of Multiplication to obtain

$$0 \times x = (0 \times x) + (0 \times x)$$

- Now $0 \times x \in \mathbb{F}$ so we can use the Existence of Additive Inverse of $0 \times x$, which we label via $-(0 \times x)$ to see that

$$0 \times x + -(0 \times x) = (0 \times x) + (0 \times x) + -(0 \times x)$$

implies

$$0 = ((0 \times x) + (0 \times x)) + -(0 \times x)$$

Now we use the Associativity of Addition to see that

$$0 = (0 \times x) + ((0 \times x) + -(0 \times x))$$

We can use the Existence of the Additive Inverse of $0 \times x$ again, and see that

$$0 = (0 \times x) + 0$$

Now we use the fact that 0 is the Additive Identity to deduce that

$$0 = 0 \times x$$

(c) The linear transformation $\alpha : \mathbb{R}^3 \to \mathbb{R}^3$ has matrix $A = \begin{pmatrix} a & b & b \\ b & a & b \\ b & b & a \end{pmatrix}$ with respect to the standard basis of $\mathbb{R}^3$, i.e. $\{e_1, e_2, e_3\}$, where $a, b \in \mathbb{R}$.
Determine the eigenvalues of A, by finding a matrix J with $A = (a-b)I + J$, or otherwise. [10 marks]
[You may want to distinguish the case $b = 0$ from the case $b \neq 0$.]

We note that

$$A = (a-b)I + bE,$$

where $(\forall i, j \in \{1, 2, 3\})E_{ij} = 1$. [1 mark]
When $b = 0$, $A = aI$ and so the eigenvalues of A are a (repeated thrice). [1 mark]
We assume that $b \neq 0$ Assume x is an eigenvector of A with eigenvalue λ, i.e. $Ax = \lambda x$. Now,

$$Ax = \lambda x \iff [(a-b)I + bE]x = \lambda x \iff Ex = \frac{[\lambda - (a-b)]}{b}x.$$

Therefore, for $b \neq 0$, x is an eigenvector of A with eigenvalue λ iff x is an eigenvector of E with eigenvalue $\frac{[\lambda-(a-b)]}{b}$. [3 marks]
We now find the eigenvalues of E. The eigenvalues of E

$$\det(E - \lambda I) = 0 \iff \lambda^2(3 - \lambda) = 0$$

[1 mark for correct characteristic equation]
Eigenvalues of E are 3 (once) and 0 (repeated twice). [2 marks]
Therefore for $b \neq 0$, the eigenvalues of A are $a + 2b$ (once) and $a - b$ (twice). We already know that when $b = 0$, the eigenvalues of A are a (repeated thrice). Therefore, for all real b, the eigenvalues of A are $a + 2b$ (once) and $a - b$ (twice).
[2 marks]

3. (20 marks)

(a) Suppose U and W are subspaces of a vector space V over the field $\mathbb{R}$. Explain what is meant by $U \cap W$ and $U + W$ and show that both of these are subspaces of V. [8 marks]

Definition of $U \cap W$: $U \cap W$ denotes the intersection of the subspaces U and W. It consists of all vectors that are in both U and W. Formally,

$$U \cap W = \{v \in V \mid v \in U \text{ and } v \in W\}$$

[1 mark]

Definition of $U + W$: $U + W$ denotes the sum of the subspaces U and W. It consists of all vectors that can be expressed as the sum of a vector from U and a vector from W. Formally,

$$U + W = \{u + w \mid u \in U, w \in W\}$$

[1 mark]

Proof that $U \cap W$ is a subspace of V :

To show that $U \cap W$ is a subspace, we verify the three subspace properties:

1. Zero Vector: Since U and W are subspaces, $0 \in U$ and $0 \in W$, so $0 \in U \cap W$.

2. Closed Under Addition: Let $v_1, v_2 \in U \cap W$. Then $v_1, v_2 \in U$ and $v_1, v_2 \in W$. Since U and W are subspaces, $v_1 + v_2 \in U$ and $v_1 + v_2 \in W$, so $v_1 + v_2 \in U \cap W$.

3. Closed Under Scalar Multiplication: Let $v \in U \cap W$ and $c \in \mathbb{R}$. Since $v \in U$ and $v \in W$, and both U and W are subspaces, $cv \in U$ and $cv \in W$, so $cv \in U \cap W$.

[3 marks]

Proof that $U + W$ is a subspace of V :

Again, we verify the three subspace properties:

1. Zero Vector: Since $0 \in U$ and $0 \in W, 0 = 0 + 0 \in U + W$.

2. Closed Under Addition: Let $v_1, v_2 \in U + W$, so $v_1 = u_1 + w_1$ and $v_2 = u_2 + w_2$ with $u_i \in U, w_i \in W$. Then $v_1 + v_2 = (u_1 + u_2) + (w_1 + w_2) \in U + W$.

3. Closed Under Scalar Multiplication: Let $v \in U + W$ and $c \in \mathbb{R}$. Then $v = u + w$ with $u \in U, w \in W$. Then $cv = cu + cw \in U + W$.

[3 marks]

(b) Let V be the vector space of functions that are square integrable (you do not have to know what square integrable means), i.e.

$$V = \left\{ f : [-\pi, \pi] \to \mathbb{R} : \int_{-\pi}^{\pi} |f(x)|^2 dx < \infty \right\}.$$

Define the following inner product:

$$(\forall f, g \in V)\langle f, g \rangle = \int_{-\pi}^{\pi} f(x)g(x)dx.$$

Consider the following set of vectors within V: $\mathcal{B} = \left\{ \begin{array}{c} \sqrt{2}\sin(x), \sqrt{2}\sin(2x), \sqrt{2}\sin(3x), \ldots, \\ \sqrt{2}\cos(x), \sqrt{2}\cos(2x), \sqrt{2}\cos(3x), \ldots, \\ 1 \end{array} \right\}$

Given that $\mathcal{B}$ is a spanning set of V, show that $\mathcal{B}$ is an orthogonal basis of V. [6 marks]

$\mathcal{B}$ is a spanning set of V iff

$$(\forall v \in V)(\exists f_i \in \mathcal{B})(\exists \lambda_i \in \mathbb{R})v = \sum_i \lambda_i f_i.$$

We need to show that $\mathcal{B}$ is a linearly independent set. Here, we exploit the fact that if two distinct elements of a set are orthogonal, the set is linearly independent. We shall now prove that

$$(\forall f, g \in \mathcal{B})f \neq g$$

we have $\langle f, g \rangle = 0$.

[1 mark]

We note that

$$\int_{-\pi}^{\pi} \sin nx dx = 0 = \int_{-\pi}^{\pi} \cos mx dx = 0 \Rightarrow \langle \sqrt{2} \sin nx, 1 \rangle = 0 = \langle \sqrt{2} \cos mx, 1 \rangle$$

[2 marks]

Since $\sin nx \cos mx$ is an odd function

$$(\forall n, m \in \mathbb{Z}^+) \int_{-\pi}^{\pi} \sin nx \cos mx = 0 \Rightarrow (\forall n, m \in \mathbb{Z}^+) \langle \sqrt{2} \sin nx, \sqrt{2} \cos mx \rangle = 0$$

[2 marks]

We have now show that

$$(\forall f, g \in \mathcal{B})f \neq g$$

we have $\langle f, g \rangle = 0$. Therefore, $\mathcal{B}$ is a linearly independent spanning set of V, and hence $\mathcal{B}$ is a basis of V. We have also show that any two distinct elements of V are orthogonal. There $\mathcal{B}$ is an orthogonal basis of V

[1 mark]

(c) Consider three uncorrelated assets, $i \in \{1, 2, 3\}$ where asset i has return variance σ_i^2. I place the fraction w_i of my wealth in asset i.

The portfolio variance is given by

$$V(w_1, w_2, w_3) = \sigma_1^2 w_1^2 + \sigma_2^2 w_2^2 + \sigma_3^2 w_3^2$$

Choose w_1, w_2, w_3 to minimize $V(w_1, w_2, w_3)$ subject to the constraint $w_1 + w_2 + w_3 = 1$. Use the method of Lagrange multipliers.

[6 marks]

We define the Lagrangian L by combining the objective function and the constraint:

$$L(w_1, w_2, w_3, \lambda) = \sigma_1^2 w_1^2 + \sigma_2^2 w_2^2 + \sigma_3^2 w_3^2 - \lambda (w_1 + w_2 + w_3 - 1)$$

[1 mark]

We take the partial derivatives of L with respect to w_1, w_2, w_3:

$(\forall i \in \{1, 2, 3\}) \frac{\partial L}{\partial w_i} = 2\sigma_i^2 w_i - \lambda$

Therefore,

$(\forall i \in \{1,2,3\})0 = 2\sigma_i^2 w_i - \lambda$

[1 mark]

Hence, $(\forall i \in \{1,2,3\})w_i = \frac{\lambda}{2\sigma_i^2}$

[1 mark]

We have the constraint $w_1 + w_2 + w_3 = 1$, and so

$$\frac{\lambda}{2}\sum_{i=1}^{3}\sigma_i^{-2} = 1.$$

[1 mark]

Therefore,

$$\lambda = \frac{2}{\sum_{i=1}^{3}\sigma_i^{-2}}.$$

[1 mark]

Hence,

$$(\forall i \in \{1,2,3\})w_i = \frac{\sigma_i^{-2}}{\sum_{i=1}^{3}\sigma_i^{-2}}.$$

[1 mark]

4. (20 marks) For $n \in \mathbb{N}$, define

$$a_n = \sum_{k=1}^{n}\frac{1}{\sqrt{n^2+k}}.$$

Decide whether or not the sequence $(a_n)_n$ converges and prove your conjecture is correct. If $(a_n)_n$ converges, find its limit. You may use standard theorems without proof provided you state where you use them.

We first write out the sequence explicitly.

$$\begin{aligned}
a_1 &= \frac{1}{\sqrt{2}} \\
a_2 &= \frac{1}{\sqrt{5}} + \frac{1}{\sqrt{6}} \\
a_3 &= \frac{1}{\sqrt{10}} + \frac{1}{\sqrt{11}} + \frac{1}{\sqrt{12}} \\
a_4 &= \frac{1}{\sqrt{17}} + \frac{1}{\sqrt{18}} + \frac{1}{\sqrt{19}} + \frac{1}{\sqrt{20}}
\end{aligned}$$

We shall bound the sequence (a_n) by two other sequences and then apply the Sequence Squeeze Theorem.

For all $n \in \mathbb{N}$, we have

$$\begin{aligned} a_n &= \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \ldots + \frac{1}{\sqrt{n^2+n}} \\ &\leq n \max\left(\frac{1}{\sqrt{n^2+1}}, \ldots, \frac{1}{\sqrt{n^2+n}}\right) \\ &= n\frac{1}{\sqrt{n^2+1}} = \frac{1}{\sqrt{1+\frac{1}{n^2}}} \end{aligned}$$

Therefore, $(\forall n \in \mathbb{N}) a_n \leq \frac{1}{\sqrt{1+\frac{1}{n^2}}}$.

[3 marks]

For all $n \in \mathbb{N}$, we have

$$\begin{aligned} a_n &= \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \ldots + \frac{1}{\sqrt{n^2+n}} \\ &\geq n \min\left(\frac{1}{\sqrt{n^2+1}}, \ldots, \frac{1}{\sqrt{n^2+n}}\right) \\ &= n\frac{1}{\sqrt{n^2+n}} = \frac{1}{\sqrt{1+\frac{1}{n}}} \end{aligned}$$

Therefore, $(\forall n \in \mathbb{N}) a_n \geq \frac{1}{\sqrt{1+\frac{1}{n}}}$.

[3 marks]

Combining our previous two results, we have

$$(\forall n \in \mathbb{N}) \frac{1}{\sqrt{1+\frac{1}{n}}} \leq a_n \leq \frac{1}{\sqrt{1+\frac{1}{n^2}}}$$

We now consider the sequence (b_n), where $b_n = \sqrt{1+\frac{1}{n}}$. Note that $(\forall n \in \mathbb{N}) b_n > 0$

Hence, we can write

$$(\forall n \in \mathbb{N}) b_n = \frac{1+\frac{1}{n}}{b_n}$$

By the Algebraic Limit Theorem

$$\lim b_n = \frac{\lim\left(1+\frac{1}{n}\right)}{\lim b_n},$$

and so

$$(\lim b_n)^2 = \lim\left(1+\frac{1}{n}\right) \Rightarrow \lim b_n = \sqrt{\lim\left(1+\frac{1}{n}\right)},$$

where we take the positive root, because $(\forall n \in \mathbb{N}) b_n > 0$

By the Algebraic Limit Theorem $\lim\left(1+\frac{1}{n}\right) = \lim 1 + \lim\frac{1}{n} = 1 + \left(\lim\frac{1}{n}\right)^2$. We now show that $\lim\frac{1}{n} = 0$.
[4 marks]
The natural numbers are unbounded

$$\begin{aligned}
&\Rightarrow (\forall\epsilon > 0)(\exists N \in \mathbb{N}) N > \frac{1}{\epsilon} \\
&\iff (\forall\epsilon > 0)(\exists N \in \mathbb{N}) \frac{1}{N} < \epsilon \\
&\Rightarrow (\forall\epsilon > 0)(\exists N \in \mathbb{N}) \left|\frac{1}{N} - 0\right| < \epsilon
\end{aligned}$$

Therefore,

$$(\forall\epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)\left|\frac{1}{n} - 0\right| < \epsilon$$

[4 marks]
Hence, $\lim\left(1+\frac{1}{n}\right) = 1$ and so $\lim b_n = 1$.

We now consider the sequence (c_n), where $c_n = \sqrt{1+\frac{1}{n^2}}$. Note that $(\forall n \in \mathbb{N}) c_n > 0$
Hence, we can write

$$(\forall n \in \mathbb{N}) c_n = \frac{1+\frac{1}{n^2}}{c_n}$$

By the Algebraic Limit Theorem

$$\lim c_n = \frac{\lim\left(1+\frac{1}{n^2}\right)}{\lim c_n},$$

and so

$$(\lim c_n)^2 = \lim\left(1+\frac{1}{n^2}\right) \Rightarrow \lim c_n = \sqrt{\lim\left(1+\frac{1}{n^2}\right)},$$

where we take the positive root, because $(\forall n \in \mathbb{N}) c_n > 0$
By the Algebraic Limit Theorem $\lim\left(1+\frac{1}{n^2}\right) = \lim 1 + \lim\frac{1}{n^2} = 1 + \left(\lim\frac{1}{n}\right)^2$. We have shown that $\lim\frac{1}{n} = 0$. Hence, $\lim\left(1+\frac{1}{n^2}\right) = 1$ and so $\lim c_n = 1$.
[4 marks]

We have now shown that

$$(\forall n \in \mathbb{N}) \frac{1}{\sqrt{1+\frac{1}{n}}} \leq a_n \leq \frac{1}{\sqrt{1+\frac{1}{n^2}}}$$

and

$$\lim \frac{1}{\sqrt{1+\frac{1}{n}}} = \lim \frac{1}{\sqrt{1+\frac{1}{n^2}}} = 1.$$

Therefore, by the Sequence Squeeze Theorem, we have
$\lim a_n = 1$.
[2 marks]

End of Paper